

Lab Report – EE-170L Computer Programming

(Lab 07)

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LAB: 7 (Programming)

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1. Objective

The main objective of this lab is to understand and practice the use of **one-dimensional arrays** in C++. Arrays are important data structures because they allow storing multiple values of the same data type in a single variable.

In this lab, we focus on:

- What is an Array
- Declaration of Arrays
- Initialization of Arrays
- Accessing Array Elements
- Copying Arrays
- Character Arrays (Strings)

By the end of this lab, students should be able to:

- Declare and initialize arrays in C++.

- Access and modify array elements using indexes.
 - Use loops with arrays for displaying and processing data.
 - Copy data from one array to another.
 - Calculate sums using arrays.
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2. Theory

2.1 What is an Array?

An **array** is a collection of elements of the same data type stored in contiguous memory locations. Arrays help programmers store and manage multiple values efficiently using a single variable name.

For example, instead of creating five separate variables, we can store all values in one array.

2.2 Declaration of Array

To declare an array in C++, we specify:

- The data type of the array
- The name of the array
- The size of the array

General Syntax:

```
dataType arrayName[arraySize];
```

Example:

```
int age[4];
```

This statement creates an integer array named age with 4 elements.

2.3 Initialization of Array

Arrays can be initialized at the time of declaration by assigning values inside curly braces.

Example:

```
int age[4] = {23, 34, 64, 75};
```

Each value is stored in the array according to its index position.

2.4 Accessing Array Elements

Array elements are accessed using indexes. In C++, array indexing starts from **0**.

Example:

```
age[0]; // First element  
age[1]; // Second element
```

Arrays are commonly accessed using loops.

3. Programs and Outputs

Task 1: Array Declaration and Initialization

```
#include <iostream>  
using namespace std;  
  
int main()  
{  
    // Declare and initialize array  
    int numbers[5] = {10, 20, 30, 40, 50};  
  
    // Loop to display array elements  
    for(int i = 0; i < 5; i++)  
    {  
        cout << numbers[i] << endl;  
    }  
  
    return 0;  
}
```

Output:

10
20
30
40
50

Program Comments:

- An integer array named numbers is created with 5 elements.
 - The array is initialized with values from 10 to 50.
 - A for loop is used to access and display each element of the array.
-

Task 2: Accessing Character Array Elements

```
#include <iostream>
using namespace std;

int main()
{
    // Declare character array
    char message[] = "HELLO";

    // Loop to access each character
    for(int i = 0; message[i] != '\0'; i++)
    {
        cout << message[i] << endl;
    }

    return 0;
}
```

Output:

H
E
L
L
O

Program Comments:

- A character array named message stores the word “HELLO”.
 - The loop continues until the null character '\0' is reached.
 - Each character is displayed on a separate line.
-

Task 3: Copying Arrays

```
#include <iostream>
using namespace std;

int main()
{
    // Declare source and destination arrays
    int source[5] = {1, 2, 3, 4, 5};
    int destination[5];

    // Copy elements using loop
    for(int i = 0; i < 5; i++)
    {
        destination[i] = source[i];
    }

    // Display source array
    cout << "Source Array: ";
    for(int i = 0; i < 5; i++)
    {
        cout << source[i] << " ";
    }

    cout << endl;

    // Display destination array
    cout << "Destination Array: ";
    for(int i = 0; i < 5; i++)
    {
        cout << destination[i] << " ";
    }

    return 0;
}
```

Output:

Source Array: 1 2 3 4 5

Destination Array: 1 2 3 4 5

Program Comments:

- Two arrays named source and destination are declared.
 - The source array is initialized with values.
 - A for loop copies each element individually into the destination array.
 - Both arrays are displayed to verify successful copying.
-

Task 4: Sum of Array Elements

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int numbers[5];
```

```
    int sum = 0;
```

```
    // Input numbers from user
```

```
    cout << "Enter 5 numbers: ";
```

```
    for(int i = 0; i < 5; i++)
```

```
    {
```

```
        cin >> numbers[i];
```

```
    }
```

```
    // Calculate sum
```

```
    for(int i = 0; i < 5; i++)
```

```
    {
```

```
        sum += numbers[i];
```

```
    }
```

```
    // Display result
```

```
    cout << "Sum = " << sum << endl;
```

```
    return 0;  
}
```

Output:

Enter 5 numbers: 1 2 3 4 5

Sum = 15

Program Comments:

- An integer array stores 5 user-entered numbers.
 - A loop is used to input the values into the array.
 - Another loop calculates the total sum of all elements.
 - The final sum is displayed on the screen.
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4. Discussion

This lab introduced the concept of arrays in C++. Arrays are very useful because they allow storing multiple values in a single variable structure.

- Arrays make programs more organized and efficient.
- Loops are commonly used with arrays to access and process data.
- Character arrays can store strings or words.
- Copying arrays requires copying each element individually using loops.

By performing these tasks, I learned how arrays work and how to use loops to manipulate array data effectively.

5. Conclusion

In this lab, I learned about one-dimensional arrays in C++. I practiced array declaration, initialization, accessing elements, copying arrays, and calculating sums using arrays.

I also learned how loops are used with arrays to process multiple values efficiently. This lab improved my understanding of arrays and strengthened my programming skills in C++

